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CryoTRANS: predicting high-resolution maps of rare conformations from self-supervised trajectories in cryo-EM



Xiao Fan^{1,10}, Qi Zhang^{2,10}, Hui Zhang⁵, Jianying Zhu¹, Lili Ju⁶, Zuoqiang Shi^{1,7}✉, Mingxu Hu^{3,4,8}✉ & Chenglong Bao^{1,7,9}✉

Cryogenic electron microscopy (cryo-EM) has revolutionized structural biology, enabling efficient determination of structures at near-atomic resolutions. However, a common challenge arises from the severe imbalance among various conformations of vitrified particles, leading to low-resolution reconstructions in rare conformations due to a lack of particle images in these quasi-stable states. We introduce CryoTRANS, a method that predicts high-resolution maps of rare conformations by constructing a self-supervised pseudo-trajectory between density maps of varying resolutions. This trajectory is represented by an ordinary differential equation parameterized by a deep neural network, ensuring retention of detailed structures from high-resolution density maps. By leveraging a single high-resolution density map, CryoTRANS significantly improves the reconstruction of rare conformations and has been validated on four real-world datasets: alpha-2-macroglobulin, actin-binding protein complexes, SARS-CoV-2 spike glycoprotein, and the 70S ribosome. CryoTRANS can also predict high-resolution structures in cryogenic electron tomography maps using a high-resolution cryo-EM map.

The elucidation of biological macromolecules' structures is fundamental to understanding life's complexities and biochemical mechanisms. Among various techniques, cryogenic electron microscopy (cryo-EM) has proven to be a powerful tool^{1–5}, enabling structural analysis at atomic or near-atomic resolutions. Remarkably, it achieves this by rapidly vitrifying biological specimens within milliseconds^{6,7}—a speed that notably surpasses the transition durations between conformational states^{8–10}.

The functions of biological macromolecules are driven by their structural transitions. According to the dynamic equilibrium law, there is a significant disparity in the prevalence of different conformational states in vitrified particles. High-occupancy states, also known as common conformations, are markedly more likely to accumulate a substantial number of particles, thereby enabling high-resolution reconstructions in comparison to less common states. Conversely, the law also suggests that high-energy quasi-stable states are sparsely populated under equilibrium conditions¹¹. Such states, named as rare conformations, hold biologists' interest due to their capacity to illuminate crucial biological processes because of their functional significance^{7,12}. However, owing to the limited availability of single-particle

images, rare conformations often yield unsatisfactory resolutions in cryo-EM when processed using the conventional discrete clustering approach^{13–15}.

Time-resolved cryo-EM^{7,10,16,17} offers a pioneering method to enhance the representation of particle images associated with rare conformations. This is achieved through meticulous control of the biochemical reaction prior to the freezing process. The innovative aspect of this technique lies in its ability to vitrify the sample precisely when the high-energy quasi-stable conformation is predominant. Successful application of time-resolved cryo-EM demands a thorough understanding of the underlying biochemical reactions, adeptness in micro-fluid mixing¹⁸, proficiency with grid preparation devices¹⁸, and seasoned expertise. An alternative strategy to address the limited occupancy of rare conformation particles is liquid cell electron microscopy (LCEM)¹⁹, which maintains the non-equilibrium conditions within the liquid cell, rather than in frozen samples. Thus, a limitation of LCEM is that, without the protective shield of cryogenic freezing, the captured images of biomolecules are of restricted resolution. Given the inherent challenges and limitations associated with these methods, there is a compelling need to craft image-processing algorithms tailored to mitigate the

¹Yau Mathematical Sciences Center, Tsinghua University, Beijing, China. ²School of Life Science, Tsinghua University, Beijing, China. ³Beijing Frontier Research Center of Biological Structure (Tsinghua University), Beijing, China. ⁴Beijing Advanced Innovation Center for Structural Biology (Tsinghua University), Beijing, China. ⁵Qiuzhen College, Tsinghua University, Beijing, China. ⁶Department of Mathematics, University of South Carolina, Columbia, SC, USA. ⁷Yanqi Lake Beijing Institute of Mathematical Sciences and Applications, Beijing, China. ⁸Institute of Bio-Architecture and Bio-Interactions, Shenzhen Medical Academy of Research and Translation, Shenzhen, China. ⁹State Key Laboratory of Membrane Biology, School of Life Sciences, Tsinghua University, Beijing, China. ¹⁰These authors contributed equally: Xiao Fan, Qi Zhang. ✉e-mail: zqshi@mail.tsinghua.edu.cn; humingxu@smart.org.cn; clbao@mail.tsinghua.edu.cn

pronounced imbalance between various conformations observed in traditional cryo-EM.

In recent years, continuous heterogeneity reconstruction methods^{13,20–25} have emerged as alternatives that may enhance the resolution of these rare conformations. These techniques extrapolate deformations directly from two-dimensional particle images. However, they suffer from an imbalance in particle distribution, which is likely to impair the effectiveness of the continuous heterogeneity reconstruction methods. An alternative approach to improving the resolution of rare conformations leverages deep learning-based post-processing techniques^{26,27}. These methods utilize neural networks trained on large datasets of cryo-EM maps to predict and refine features in low-resolution input maps. However, it is important to note that the performance of such methods heavily depends on the pre-existing structural information in the training datasets, which may introduce model bias. This bias can result in incorrect modeling for structures that are underrepresented in the training data²⁷, thereby affecting the reliability of the enhanced maps.

In this paper, we introduce a method called CryoTRANS (Transformations by Artificial Networks). Our method accurately predicts high-resolution density maps of rare conformations by identifying a pseudo-trajectory through different states in a self-supervised manner. CryoTRANS employs a deep neural network-parameterized ordinary differential equation (ODE), termed neural ODE, to model the continuum of non-rigid deformations from common to rare conformations. By incorporating a Wasserstein distance and imposing a rigidity constraint on the velocity field, we propose an optimization model for training the neural network in a self-supervised manner, eliminating the need for supervised samples or pre-

training. Upon training completion, the deformed density map, governed by the ODE, yields a high-resolution approximation of the rare conformations. We demonstrate the efficacy of CryoTRANS through four real datasets, validating its capability to predict high-resolution density maps of rare conformations with high accuracy. We further demonstrate the advantages of CryoTRANS over the existing deep-learning-based post-processing methods, including DeepEMhancer²⁶ and EMReady²⁷. In addition, we compare the pseudo-trajectories generated by CryoTRANS with MorphOT²⁸ and CryoDRGN²², showing that our trajectory does not suffer from the random appearance and disappearance of densities during the deformation process. In particular, the pseudo-trajectory generated by CryoTRANS exhibits consistency with molecular dynamics simulations, further validating the robustness and accuracy of our approach.

Results

Overview of CryoTRANS

CryoTRANS enables morphing between two cryo-EM density maps using an ODE-governed deformation field parameterized by a deep neural network (DNN) (Fig. 1a). In this process, the initial state f_0 corresponds to a high-resolution density map, while the terminal state f_1 corresponds to a low-resolution target map. The deformation process, represented by a velocity field parameterized by the network $\vec{V}$, is captured by varying the pseudo-time in the interval $[0, 1]$. The discretization of the ODE model in time naturally results in a recurrent weight-sharing network capable of mapping any point in the initial density map to a corresponding point in $\mathbb{R}^3$. Given $t \in [0, 1]$, we define $\hat{f}_t$ as the intermediate map at time t (Fig. 1c).

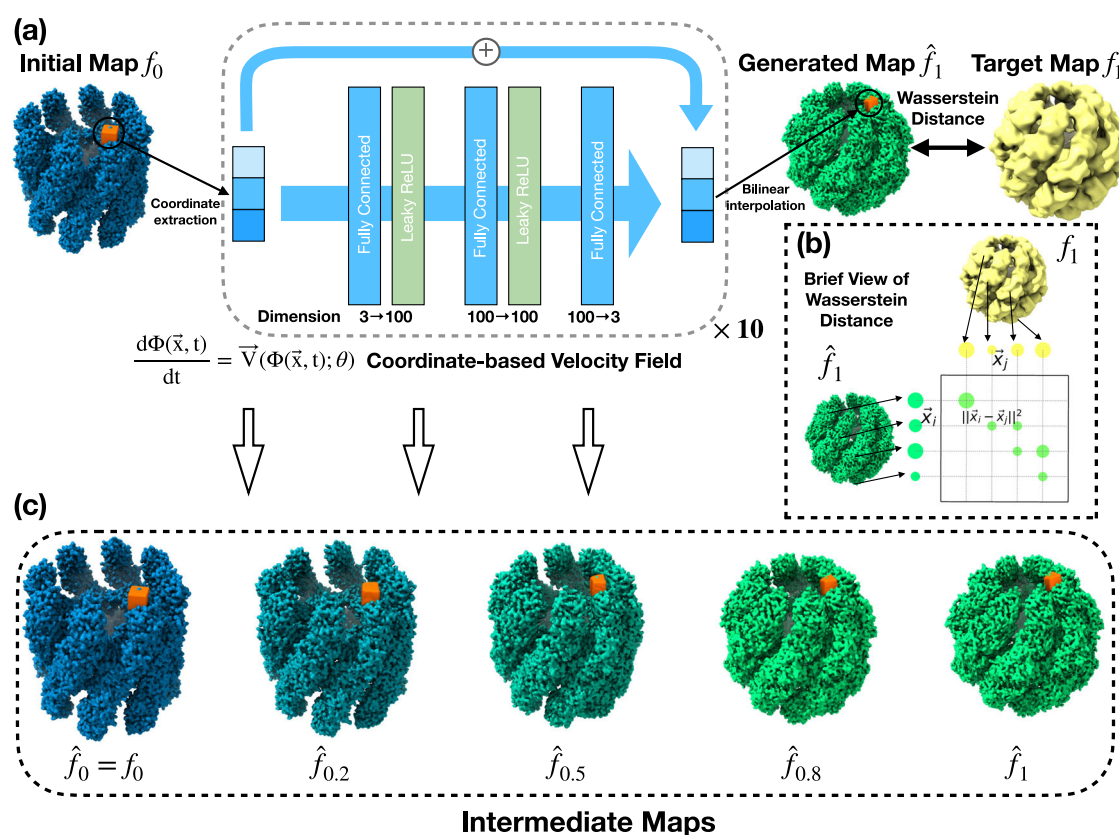


Fig. 1 | The introduction of CryoTRANS. **a** CryoTRANS leverages a high-resolution initial map denoted by f_0 and a lower-resolution target map denoted by f_1 as inputs. It then utilizes a neural network-parameterized velocity field $\vec{V}$ to construct a pseudo-trajectory within the 3D coordinate space of the initial map. This pseudo-trajectory is achieved by discretizing a governing ODE into ten pseudo-time points. During training, the model minimizes a loss function that incorporates the Wasserstein distance between the final generated map $\hat{f}_1$ and the target map f_1 . The gradient of this loss function is backpropagated through the neural network to

optimize its parameters. **b** Diagram of the Wasserstein Distance: quantifies the minimum effort required to transform the densities in the generated map $\hat{f}_1$ to match those in the target map f_1 . **c** The intermediate density maps at different pseudo-time points generated by CryoTRANS using the trained velocity field $\vec{V}$. This pseudo-trajectory, visualized as an orange cube, represents the path of a specific voxel within the density map. These example maps are simulations derived from atomic models of Mm-cpn^{34,35} in two conformations: the closed state (PDB 3J03) and the open state (PDB 3IYF).

Specifically, the initial map $\hat{f}_0$ is equal to f_0 , and the terminal map $\hat{f}_1$ is called as the generated map (Fig. 1a).

To train the DNN, the squared Wasserstein distance is employed as the loss function between the generated map $\hat{f}_1$ and the target map f_1 , opting for it over the conventional mean squared error. The Wasserstein distance, which is the minimal transportation cost between two probability distributions, effectively captures the geometric information of the underlying space. Specifically, let $\vec{x}_i$ be location of the i th voxel in $\hat{f}_1$ and $\vec{x}_j$ be location of the j th voxel in f_1 , $\|\vec{x}_i - \vec{x}_j\|^2$ (squared Euclidean norm) and P_{ij} represent the cost and the probability density moving from $\vec{x}_i$ to $\vec{x}_j$, respectively. Then $P_{ij} \|\vec{x}_i - \vec{x}_j\|^2$ is the transportation cost that relocates from $\vec{x}_i$ in $\hat{f}_1$ to $\vec{x}_j$ in f_1 (Fig. 1b). The squared Wasserstein distance corresponds to the minimal total transportation cost, which is the summation of transportation costs for all voxels, achieved by finding a suitable transport plan P . Recent studies have shown the advantages of Wasserstein distance over classical Euclidean distances with theoretical analysis²⁹, which validates the mathematical soundness of CryoTRANS. Additionally, a rigidity constraint is placed on the velocity field $\vec{V}$ to preserve the mass of the density map in each layer and prevent the dispersion of density. The network is optimized using the adaptive moment estimation (ADAM) algorithm³⁰, and we provide a more detailed exposition of CryoTRANS in the “Methods” section.

Within the trained velocity field $\vec{V}$, it is observed that the high-resolution details of the initial map are maintained throughout the deformation process (Fig. 1c). Thus, the generated map can be seen as a high-resolution prediction of the target map, effectively capturing the structural features.

Validation of the CryoTRANS-generated maps

To evaluate the quality and accuracy of the CryoTRANS-generated maps, we conducted experiments on simulated datasets of four biological macromolecules. For each macromolecule, two distinct conformations have been resolved in previous studies using experimental approaches, and corresponding atomic models have been constructed. Specifically, the four are the heterodimeric ABC exporter TmrAB³¹ in ATP-bound outward-facing open conformation (PDB 6RAH) and ATP-bound outward-facing occluded conformation (PDB 6RAI), another ABC exporter PCAT1³² in the inward-facing wide conformation (PDB 7T55) and the inward-facing narrow conformation (PDB 7T57), human prestin³³ in complex with chloride (PDB 7V73) and salicylate (PDB 7V75), and Mm-cpn^{34,35} in its open state (PDB 3IYF) and closed state (PDB 3J03). To mimic target maps with different resolutions, we low-pass filtered the atomic model of one conformation state to 3, 5, and 7 Å. We compared the generated maps to the atomic models and calculated the model-to-map Fourier shell correlation (FSC) (Fig. 2). Additional quantitative metrics, including Q-score³⁶, TM-score³⁷, and chain comparison score of Phenix³⁸, can be found in Supplementary Note I.

If both the initial and target density maps are at 3 Å resolution, the generated maps have a resolution higher than 3.2 Å (Fig. 2). Additionally, when the target maps are at 5 or 7 Å resolution, the generated maps preserve information up to an approximate resolution of 4 Å (Fig. 2). These results demonstrate that the density maps generated by CryoTRANS can preserve the correct structural details using the learned velocity field.

Furthermore, we explored the performance of CryoTRANS when simultaneously morphing multiple different target conformations based on the simulated data of TmrAB (Supplementary Note II). The results indicated that incorporating constraints from an intermediate conformation can further improve the quality of the CryoTRANS-generated density for the target map.

CryoTRANS predicts high-resolution maps for rare conformations

We validated the effectiveness of CryoTRANS by applying it to three real-world cryo-EM datasets: the alpha-2-macroglobulin (A2M)³⁹ (Fig. 3a), a component of the actin-binding protein complexes (Arp2/3)⁴⁰ (Fig. 3f) and the 70S ribosome⁴¹ (Supplementary Note III). In these cases, the original atomic models, constructed from low-resolution target maps, inherently contained uncertainties and inaccuracies. To address these uncertainties

and validate the quality of our generated map, we refined the target model using our generated map through real space refinement in Phenix³⁸. The root mean square deviation (RMSD) was kept below 3 Å, and the maximum deviation (calculated as the infinity norm of the averaged error vector between all atoms) was restricted to <0.5 Å (Fig. 3d, g). Subsequently, the model-to-map FSC of the generated map with the refined atomic model was computed (Fig. 3e, h). These experiments demonstrate that CryoTRANS produces a density map that accurately reflects the high-resolution structures in the target conformation.

A2M, a protease inhibitor comprised of four identical subunits, exhibits both an induced state (EMDB-32052) and a native state (EMDB-32051), reported to have half-map resolutions of 3.8 and 5.3 Å, respectively³⁹. The 5.3 Å resolution of native A2M poses challenges in atomic model building. Utilizing the induced state as the starting map (model-to-map FSC of 4.32 Å) and the native state as the target map (model-to-map FSC of 8.02 Å), CryoTRANS generated a high-quality pseudo-trajectory (Fig. 3a). The refined target model's model-to-map FSC (PDB 7VON) with the generated map achieved a resolution of 4.84 Å. Comparative visualization of the side chains of the generated (in green) and target (in yellow) maps is shown in Fig. 3b, with the atomic model integrated into these density maps. This demonstrates that the generated map maintains the initial map's high-resolution features and effectively approximates the native state.

We applied CryoTRANS to Arp2/3, a component of the actin-binding protein complexes. Due to its flexibility, the target conformation (EMDB-10960) exhibits a section of missing density. By applying the CryoTRANS-generated deformation to the initial map (Fig. 3d), the missing density in the target conformation is seamlessly supplemented, revealing many high-resolution details. The model-to-map FSC between the generated map and the refined atomic model (PDB 6YW7) maintains a resolution of 3.79 Å, comparable to the initial map's resolution of 3.72 Å and significantly higher than the target map's 9.82 Å resolution (Fig. 3e). These results validate our model's ability to generate high-resolution approximations for rare conformations, even in cases where the target exhibits missing density sections.

To further validate the effectiveness of CryoTRANS, we compared it with post-processing methods based on deep learning, including DeepEMhancer²⁶ and EMReady²⁷, which directly enhance the resolution of target conformations (Fig. 3c, i). For a fair comparison, we applied generated densities from DeepEMhancer and EMReady to refine the target model, respectively, and calculated the model-to-map FSC relative to the refined model (Fig. 3e, h). The results show that CryoTRANS significantly outperforms these methods in terms of the resolution of the generated maps. It is worth noting that CryoTRANS is a self-supervised method, which does not require pretraining on large datasets and does not suffer from model bias risks.

CryoTRANS predicts high-resolution details for cryo-ET map

CryoTRANS demonstrates the potential in bridging cryo-EM single-particle analysis (SPA) and cryogenic electron tomography (cryo-ET). By utilizing a high-resolution SPA map as the starting point, CryoTRANS can effectively predict high-resolution structures from low-resolution cryo-ET maps. In the case of the SARS-CoV-2 spike glycoprotein (Fig. 4a), the initial map presents three receptor binding domains (RBDs) in a closed state at 3.57 Å (model-to-map FSC) from cryo-EM SPA (EMDB-22001), and the target map depicts one RBD in an open state at 13.2 Å (model-to-map FSC) from subtomogram averaging of cryo-ET (EMDB-16697, PDB 6ZGG). CryoTRANS successfully generates a rotation of the RBD domain while maintaining a high-resolution structure (Fig. 4b). The model-to-map FSC resolution (with the refined model) of the generated density map by CryoTRANS reaches 3.78 Å (Fig. 4c), surpassing the resolutions achieved by deep learning post-processing strategies EMReady and DeepEMhancer, which are 9.62 and 9.34 Å, respectively. Furthermore, to validate the accuracy of the CryoTRANS-generated map, we compared it with the SPA map (EMDB-21999) using the corresponding model (PDB 6X2A) as the reference. The

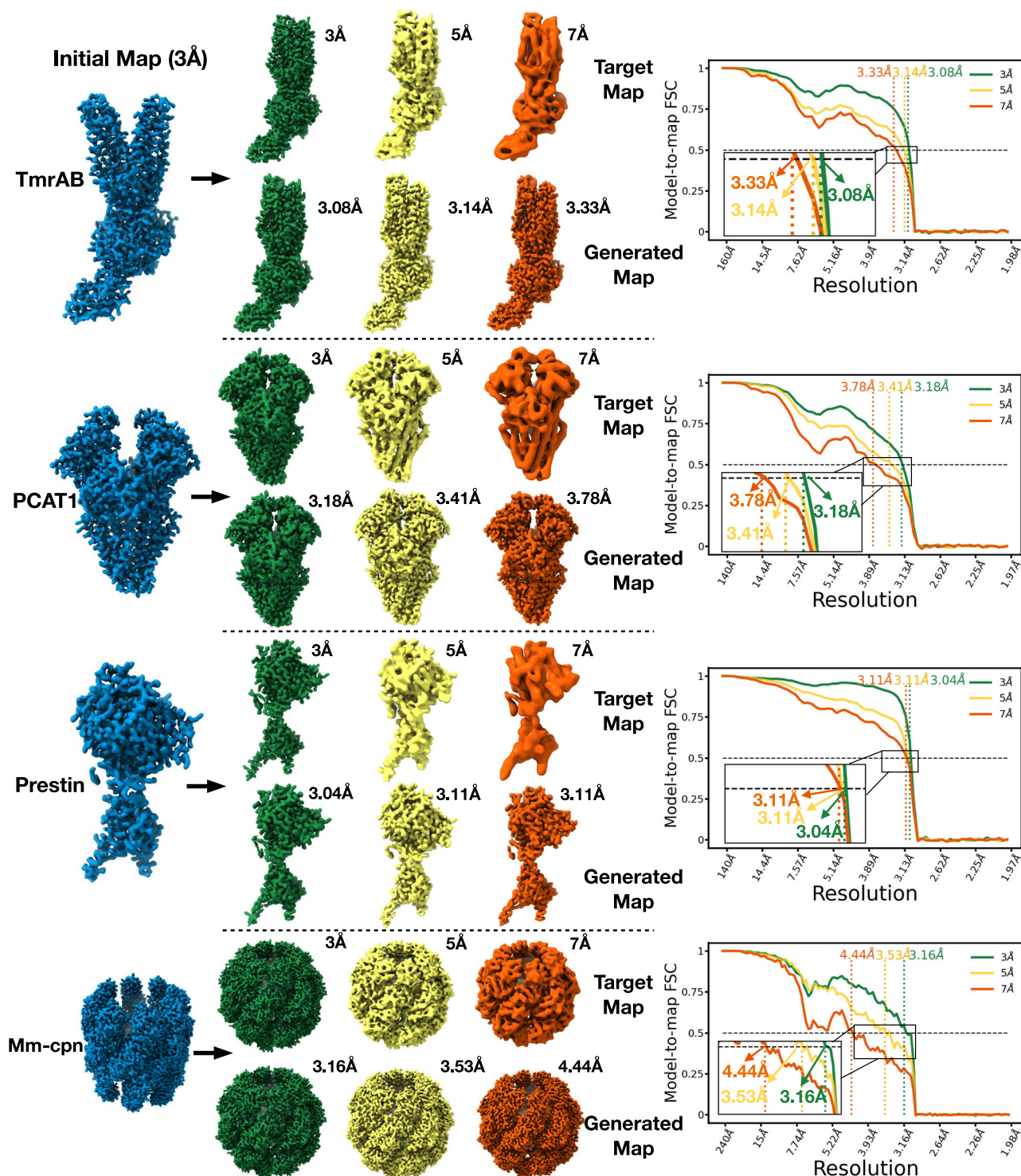


Fig. 2 | Prediction accuracy of CryoTRANS. The four biological macromolecules tested by CryoTRANS are shown from top to bottom. Each macromolecule exists in two distinct conformations, with atomic models of these conformations resolved using experimental methods. The left column shows the initial maps for CryoTRANS, generated from one conformation of each protein at a resolution of 3 Å. The middle

columns display the target maps (upper row) and the corresponding generated maps by CryoTRANS (lower row). The target maps were generated at resolutions of 3 Å (green), 5 Å (yellow), and 7 Å (orange). The right column presents the model-to-map FSCs between the ground truth atomic model and the generated maps.

computed model-to-map FSC resolution of CryoTRANS-generated density and the SPA structure was both 3.78 Å (Fig. 4d). This validation underscores CryoTRANS's ability to effectively leverage the structure obtained in SPA to enhance cryo-ET density resolution, offering valuable insights for cross-modal fusion of different imaging modalities.

The comparison between pseudo-trajectories generated from CryoTRANS and other methods

We first compared CryoTRANS with MorphOT²⁸, another morphing method for two cryo-EM density maps, using two selected cases for this comparison. The first is the simulated data derived from Mm-cpn^{34,35} in its open state (PDB 3IYF) and closed state (PDB 3J03). The second example

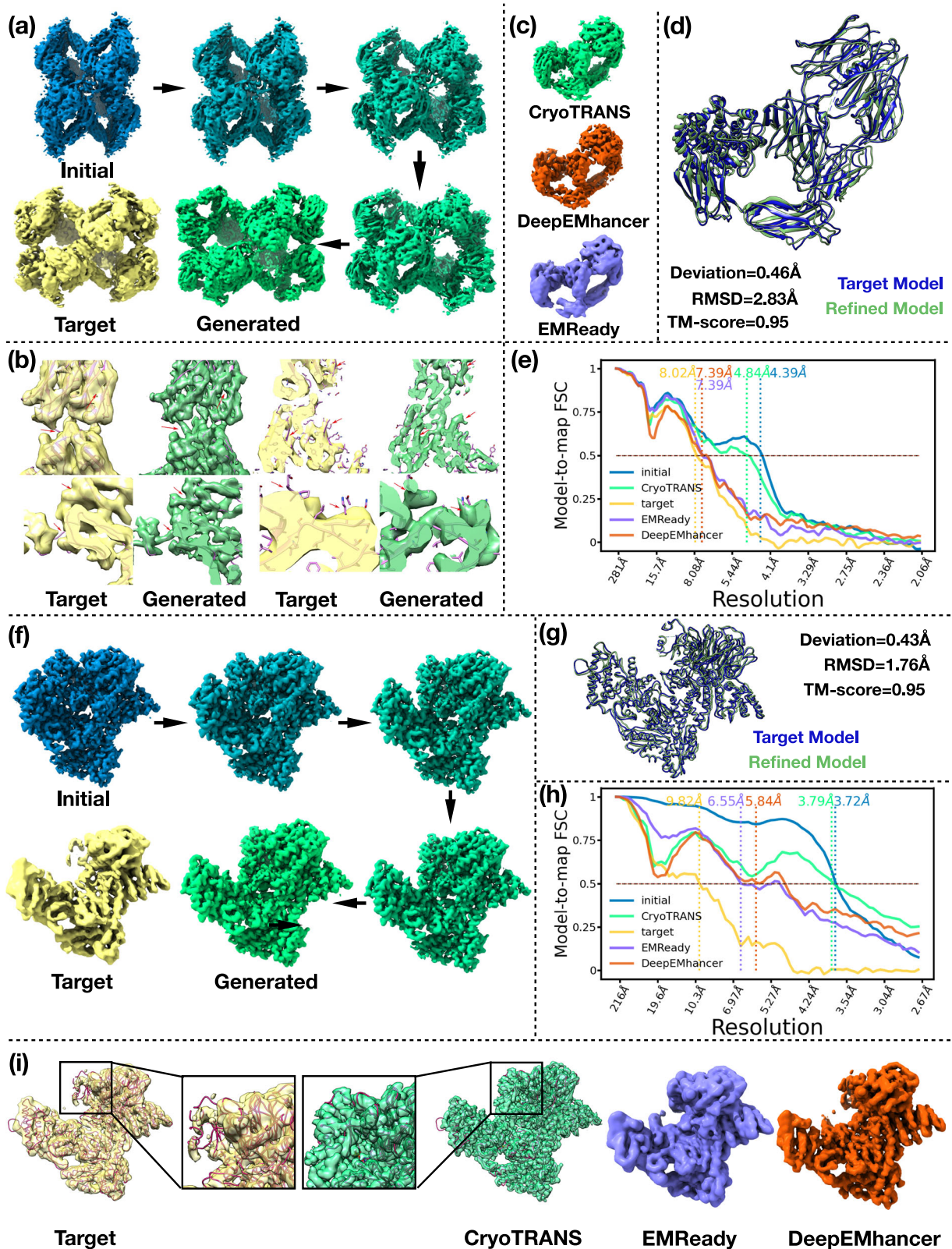


Fig. 3 | CryoTRANS predicts high-resolution maps of rare conformations in A2M and Arp2/3. a CryoTRANS analysis of A2M from the induced state (EMDB-32052, blue) to the native state (EMDB-32051, yellow). **b** Side chains of the generated map (green) and the target map (yellow) for A2M. **c** Density of an A2M subunit obtained by CryoTRANS, EMReady, and DeepEMhancer. **d** The target model (PDB 7VON) of one subunit of A2M before and after structural refinement is super-imposed for depiction, labeled with various metrics of similarity. **e** The model-to-map FSC curves for the initial map (with the induced state atomic model), generated

maps via various methods, and the target map (with the refined native state atomic model) of A2M. **f** Similar to **a**, but the data analysis involves two conformations of Arp2/3, with a home-collected initial map and EMDB-10960 as the target map. **g** Similar to **d**, but the atomic model (PDB 6YW7) corresponds to the target map of Arp2/3. **h** Similar to **e**, but for the analysis of Arp2/3. **i** Atomic models of Arp2/3 are fitted to the target map and the generated map, respectively, and compared with maps generated by EMReady and DeepEMhancer.

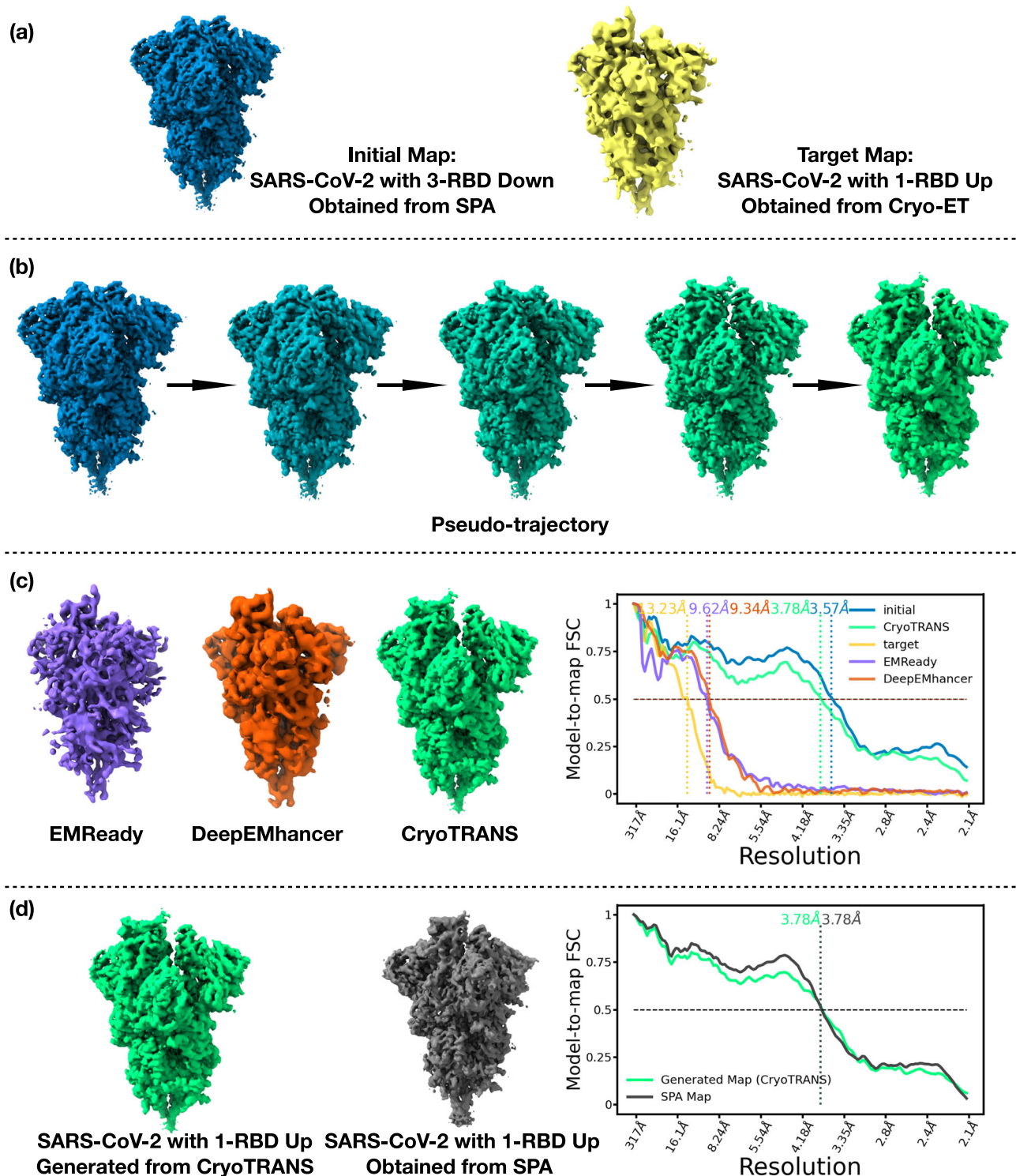


Fig. 4 | CryoTRANS uses an SPA map to obtain a high-resolution approximation for a SARS-CoV-2 cryo-ET map. a The initial map is the SARS-CoV-2 spike protein with three RBDs down obtained from cryo-EM SPA (EMDB-22001, model-to-map FSC resolution is 3.57 Å) and the target map is chosen as the SARS-CoV-2 spike protein one RBD up (EMDB-16697, model-to-map FSC resolution is 13.2 Å) obtained from cryo-ET subtomogram averaging. **b** The CryoTRANS pseudo-trajectory between the SPA map and cryo-ET map is depicted by selecting five intermediate density maps along the trajectory. **c** The model-to-map FSC curves for

the initial map (with respect to the corresponding model PDB 6X2C), generated map (with respect to PDB 6ZGG after structure refinement), and target map of SARS-CoV-2 are compared, along with the generated results from EMReady and DeepEMhancer (with respect to PDB 6ZGG after structure refinement). **d** The panel shows the density maps from CryoTRANS and cryo-EM SPA (left), and their corresponding model-to-map FSC curves (right), comparing the maps with the model (PDB 6X2A). The SPA map is of the SARS-CoV-2 spike protein with one RBD up (EMDB-21999, model-to-map FSC resolution is 3.78 Å).

involves two distant conformations of the SARS-CoV-2 spike glycoprotein (EMDB-22001, EMD-21999)⁴². In the Mm-cpn simulated dataset, the inherent constraints of MorphOT's regularization process in solving optimal transport result in indistinct interpolations at coarser resolutions. Such a limitation hinders the preservation of intricate structural nuances throughout interpolation. Conversely, our methodology adeptly avoids this issue, permitting an elevated degree of accuracy and consistent retention of structural details (Supplementary Fig. 1a). The commendable efficacy of our approach accentuates its utility as a potent instrument for morphing cryo-EM density maps. When considering the real-world scenario involving the SARS-CoV-2 spike glycoprotein, the performance disparity between MorphOT and CryoTRANS trajectories echoes the patterns identified in the Mm-cpn case. Besides, there is a noticeable distortion in the intermediary density maps from MorphOT (Supplementary Fig. 1b).

We next compared CryoTRANS with two representative continuous heterogeneity reconstruction methods, CryoDRGN²² and 3DFlex²⁰. CryoDRGN predicts a density-map trajectory by traversing a linear path within the latent space. The training of CryoDRGN involves two-dimensional particle images, while CryoTRANS relies solely on the initial and target maps. We compared the trajectories generated from two methods using a pre-catalytic spliceosome dataset (EMPIAR-10180). Specifically, we first selected two density maps from the *K*-means clusters²² generated in the CryoDRGN latent space as the initial and target maps (Fig. 5a). Using these two density maps, we generated a linear path in the latent space as the trajectory from CryoDRGN and applied CryoTRANS to align these two maps for comparison (comparison with other trajectories can be found in Supplementary Note IV). The intermediate states of the two trajectories did not match. The trajectory generated by CryoDRGN was unrealistic, with densities emerging and disappearing sporadically, deviating from basic physical constraints. In contrast, CryoTRANS effectively captured the relative motion of the SF3b component (Fig. 5b). Furthermore, our analysis with the simulated TmrAB dataset indicated that CryoDRGN tends to struggle with the imbalance of particle distribution across conformations; notably, it fails to observe the rare conformations whereas CryoTRANS effectively overcomes this issue (Supplementary Note V).

3DFlex is another deep learning-based algorithm that predicts protein motion from the learned latent space. We conducted a comparative experiment with 3DFlex, also using simulated particles of TmrAB (Supplementary Note V). In this experiment, the high-resolution initial map and the low-resolution target map for training CryoTRANS were reconstructed from a simulated dataset with an imbalanced distribution of particles corresponding to each conformation. The trajectory produced by 3DFlex, trained on all simulated particles, was then compared with CryoTRANS (Supplementary Note V). The results revealed that the 3DFlex software's default trajectory predominantly produced local perturbations of the initial map. We manually chose a trajectory that yielded a target map close to the occluded conformation. However, this density map is still inferior to the one obtained from CryoTRANS, as measured by the Q-score and residue distances.

To further validate the accuracy of the CryoTRANS pseudo-trajectory, we compared it with the trajectory generated by molecular dynamics (MD) for SARS-CoV-2⁴³ (Supplementary Note VI). The MD trajectory investigates the conformational changes of SARS-CoV-2 in the receptor-binding domain (RBD) at spaced time intervals from 0 to 180 nanoseconds (ns). The first frame of the MD trajectory ($t = 0$ ns) was used as the initial map, and the last frame ($t = 180$ ns) was used as the target map. We then generated the trajectory between these two maps using CryoTRANS. The results showed that while there are some discrepancies in the temporal distribution between the two methods, the overall trend of deformation remains consistent. After adjusting the true timeline (0 to 180 ns) in MD to the pseudo-timeline ($[0, 1]$) of CryoTRANS, the CryoTRANS pseudo-trajectory closely matches the MD trajectory. The error (3–5 Å) may largely be due to the Brownian motions in MD simulations, which introduce thermal variations that CryoTRANS cannot accurately capture. Previous studies have shown that the Brownian motion of proteins in water is on the order of Angstroms⁴⁴. It is important to note that Brownian motion is inherent to the simulation and is accounted for in the dynamics of the system, which can

vary depending on the specific conditions of the simulation. Furthermore, it is worth mentioning that these findings are specific to the studied case, and further investigation is needed to make broader generalizations.

Discussion

In this study, we introduced a novel method named CryoTRANS to predict high resolutions for rare conformations in cryo-EM studies by generating a pseudo-trajectory across various conformations. CryoTRANS leverages an ODE, parameterized by a deep neural network, to model a continuous, non-rigid deformation from common high-resolution conformations to rare ones. The method also features the novel incorporation of a Wasserstein distance and the application of a rigidity constraint on the velocity field, together forming a variational model. Mathematically, the Wasserstein distance provides a metric in the probability space, whereas the conventional Euclidean distance does not. Theoretical advantages of the Wasserstein distance over the Euclidean distance have been studied previously^{29,45}, demonstrating success in machine learning⁴⁶ and seismic data processing⁴⁷. The rigidity constraint helps in preserving the detailed structure of the high-resolution density map. This design obviates the need for supervised samples or pre-training during the neural network training process. After training, the deformed density map, governed by the ODE, offers a high-resolution counterpart of the rare conformations. The potential and effectiveness of CryoTRANS have been demonstrated through its application in various case studies, where it succeeded in predicting high-resolution features of rare conformations and cryo-ET maps.

The exploration of rare conformations in structural biology holds significant value, as these rare states often yield crucial insights into the function and behavior of biological macromolecules. However, the infrequent occurrence of these rare conformations presents a challenge in their analysis. This is where deformation models prove to be instrumental. The utilization of deformation models to generate intricate, high-resolution trajectories of macromolecules, especially from multi-modal data, holds promise in other applications as well. For instance, particle data obtained from liquid-phase electron microscopy⁴⁸ contain valuable temporal information. However, reconstructing high-resolution movies from liquid-phase EM is still an unsolved challenge, primarily due to constraints in the number of available particles and the high level of noise. Our future endeavors will focus on exploring CryoTRANS as an adaptable solution to address and overcome this ongoing issue. In conclusion, CryoTRANS serves as a tool for exploring rare conformations in structural biology by utilizing neural network deformation models. This approach may open up promising avenues for gaining deeper insights into the function and behavior of biological macromolecules.

CryoTRANS is a predictive approach, although it is self-supervised and does not require labeled data or pretraining. While CryoTRANS has shown successful results in various cases, the generated maps and pseudo-trajectories may be influenced by the choice of loss function and need further validation from both experimental and computational perspectives. In the future, more interactions between image registration-based methods and molecular dynamics simulations could provide cross-validation in computations. Additionally, using the pseudo-trajectory generated by CryoTRANS to identify particles belonging to rare conformations—typically overlooked by conventional classification methods due to imbalance clustering issues—deserves future work.

Methods

Generation of simulated datasets

Simulated density maps with varying resolutions were generated, and morphing from the initial map to the simulated density maps was performed using CryoTRANS. Four biological macromolecules were selected for generating simulated datasets, each containing two conformations resolved by experimental approaches. These macromolecules include the heterodimeric ABC exporter TmrAB (PDB 6RAH and PDB 6RAI), the peptidase-containing ABC transporter 1 (PDB 7T55 and PDB 7T57), thermostabilized human prestin (PDB 7V73 and PDB 7V75), and Mm-cpn

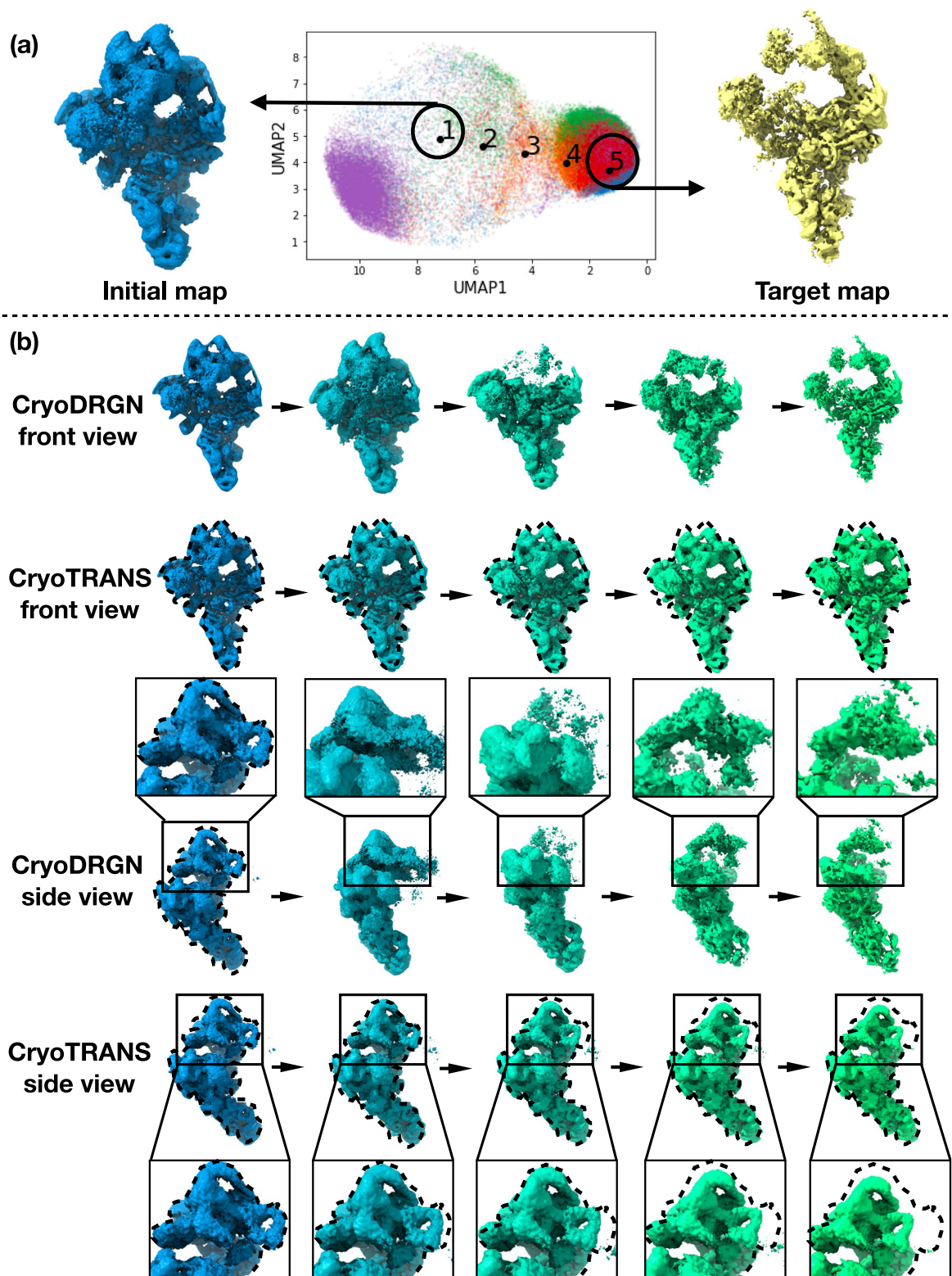


Fig. 5 | Comparison of CryoTRANS and CryoDRGN pseudo-trajectories using a pre-catalytic spliceosome dataset. a The UMAP visualization of latent encodings (middle) after training a 10-dimensional latent variable with CryoDRGN on the spliceosome dataset (EMPIAR-10180). Five positions are pinpointed along a linear path (the trajectory of CryoDRGN) within the latent space, each representing a density map encoded by the VAE's decoder within CryoDRGN. The initial and

target maps are defined as the density maps at the starting and ending points of CryoDRGN's trajectory, respectively. **b** The trajectories of CryoDRGN and CryoTRANS are juxtaposed for comparison, considering both front and side perspectives. For a clearer visualization of the movements, the contour of the density map is also displayed. A local magnification of the side view is showcased through a highlighted frame.

(PDB 3IYF and PDB 3J03). The corresponding atomic model of the simulated dataset was converted to the density map using the *molmap* function of Chimera. The density map of one conformation was low-passed to 3 Å, serving as the initial high-quality map for morphing using CryoTRANS. The density map of the other conformation was treated as the target density map and was low-passed to different resolutions of 3, 5, and 7 Å, respectively.

Mathematical principles of CryoTRANS

Basic formulation of CryoTRANS self-supervised training. CryoTRANS trains a flow map, $\vec{\Phi}(\vec{x}, t) \in \mathbb{R}^3$, $t \in [0, 1]$, in a self-supervised manner between the high-resolution density map f_0 and the low-resolution density map f_1 . The high-resolution prediction of f_1 is given by $f_0(\vec{\Phi}^{-1}(\cdot, 1))$. This flow map, $\vec{\Phi}(\vec{x}, t)$, is determined by an ODE model:

$$\begin{aligned} \frac{d\vec{\Phi}(\vec{x}, t)}{dt} &= \vec{V}(\vec{\Phi}(\vec{x}, t)), t \in [0, 1] \\ \vec{\Phi}(\vec{x}, 0) &= \vec{x}. \end{aligned} \quad (1)$$

The velocity field $\vec{V}$ is formulated as a multi-layer perceptron (MLP) with two hidden layers:

$$\vec{V}(\vec{x}; \theta) = W_3(\sigma(W_2(\sigma(W_1\vec{x} + b_1)) + b_2)) + b_3, \quad (2)$$

where σ is the activation function, and $W_1 \in \mathbb{R}^{n \times 3}$, $b_1 \in \mathbb{R}^n$, $W_2 \in \mathbb{R}^{n \times n}$, $b_2 \in \mathbb{R}^n$, $W_3 \in \mathbb{R}^{3 \times n}$, and $b_3 \in \mathbb{R}^3$ are learnable parameters. Here, θ denotes all the learnable parameters. In this study, the width of the hidden layer, n , is set to 100 by default, and σ is chosen as the leaky ReLU function with a slope of 0.01.

To learn the velocity field, the loss is defined as the squared Wasserstein distance between $f_0(\vec{\Phi}^{-1}(\cdot, 1))$ and f_1 . The high-resolution density map is obtained by solving the following optimization model:

$$\begin{aligned} \min_{\theta} \mathcal{W}_2(f_1, f_1)^2 \\ \text{subject to } \frac{d\vec{\Phi}(\vec{x}, t)}{dt} &= \vec{V}(\vec{\Phi}(\vec{x}, t); \theta), t \in [0, 1] \\ \vec{\Phi}(\vec{x}, 0) &= \vec{x}, \\ \hat{f}_1(\vec{x}) &= f_0(\vec{\Phi}^{-1}(\vec{x}, 1)). \end{aligned} \quad (3)$$

Here, $\mathcal{W}_2$ is the squared Wasserstein distance⁴⁹. After solving this optimization problem, the velocity field $\vec{V}$ and the corresponding flow $\vec{\Phi}(\vec{x}, t)$ can be obtained by solving Eq. (1).

Discretization of the neural ODE. The optimization of the neural ODE, as described in Eq. (1), requires discretization before it can be computed. Without loss of generality, the computational domain is chosen as $[0, 1]^3$ and is discretized using uniform grids:

$$\vec{x}_{ijk} = \left(\frac{i}{N}, \frac{j}{N}, \frac{k}{N} \right), \quad i, j, k \in \{0, 1, \dots, N-1\}. \quad (4)$$

The ODE (1) is discretized using the forward Euler method:

$$\begin{aligned} \vec{\Phi}(\vec{x}_{ijk}, t_{m+1}) &= \vec{\Phi}(\vec{x}_{ijk}, t_m) + \tau \vec{V}(\vec{\Phi}(\vec{x}_{ijk}, t_m); \theta), \\ \vec{\Phi}(\vec{x}_{ijk}, 0) &= \vec{x}_{ijk}, \end{aligned} \quad (5)$$

where $t_m = \frac{m}{M}$, $m \in \{0, 1, \dots, M-1\}$ and $\tau = \frac{1}{M}$. A typical choice for M is 10. Then $\hat{f}_1$ can be obtained as follows:

$$\hat{f}_1(\vec{\Phi}(\vec{x}_{ijk}, 1)) = f_0(\vec{x}_{ijk}). \quad (6)$$

In the computation, the target density map f_1 is evaluated on the uniform grids $\vec{x}_{ijk}$. To compute $\mathcal{W}_2(f_1, f_1)$, $\hat{f}_1$ also needs to be evaluated over $\vec{x}_{ijk}$. This is done using bilinear interpolation (see Supplementary Note VII).

Implementation of CryoTRANS training. The Wasserstein distance $\mathcal{W}_2(\hat{f}_1, f_1)$ is computed using the Sinkhorn algorithm⁵⁰, which is an efficient, GPU-friendly algorithm that employs entropic regularization and allows for efficient gradient backpropagation (detailed in Supplementary Note VIII). To alleviate computational overhead, the first 5 iterations of the Sinkhorn algorithm are performed in each round, and the penalty coefficient of the entropic regularization is set to 0.0001.

The gradient is computed based on automatic gradient propagation in PyTorch (see Supplementary Note VII). The weights of the neural network are initialized following a normal distribution with a mean of zero and a variance of 0.01. The model is trained using the ADAM optimizer with a learning rate of 0.001. A multi-scale algorithm is also introduced during the training process, achieving acceleration by a factor of 5 to 10 across different datasets (Supplementary Note IX).

Pseudo-trajectory inference after CryoTRANS training. Once a neural velocity field is trained with optimal parameters θ^* , (5) is further iterated with θ^* to obtain an optimal moving trajectory for each voxel $\vec{x}_{ijk}$. All the trajectories obtained in this process are gathered and bilinear interpolation is utilized to generate the final trajectory of the density map:

$$\hat{f}_{t_m} = f_0(\vec{\Phi}^{-1}(\cdot, t_m)), \quad m = 0, 1, \dots, M. \quad (7)$$

The density map $\hat{f}_1$ corresponds to the generated density map at the end of the trajectory.

Density maps partition strategy with rigidity constraints during CryoTRANS training

The procedure described above is utilized for training simulated data. However, when training CryoTRANS with real data (A2M, Arp2/3, SARS-CoV-2), where different domains follow different velocity fields, a modified approach is employed. This approach utilizes two velocity fields with rigidity constraints based on a manual partition of the density map, enhancing the model's representation capability.

Specifically, the whole volume is manually divided into two disjoint parts,

$$\Omega = [0, 1]^3 = \Omega_a \sqcup \Omega_b, \quad (8)$$

and f_0 is consequently divided into f_0^a and f_0^b , where

$$f_0^a(\vec{x}) = \begin{cases} f_0(\vec{x}), & \text{if } \vec{x} \in \Omega_a \\ 0, & \text{if } \vec{x} \in \Omega_b \end{cases}, \quad f_0^b(\vec{x}) = \begin{cases} 0, & \text{if } \vec{x} \in \Omega_a \\ f_0(\vec{x}), & \text{if } \vec{x} \in \Omega_b \end{cases}. \quad (9)$$

Subsequently, two separate neural velocity fields, characterized by different parameters θ^a and θ^b , are employed to generate trajectories on these individual partitions. The trajectory for each partition can be described as follows:

$$\begin{aligned} \vec{\Phi}^a(\vec{x}_{ijk}, t_{m+1}) &= \vec{\Phi}^a(\vec{x}_{ijk}, t_m) \\ &+ \tau \vec{V}(\vec{\Phi}^a(\vec{x}_{ijk}, t_m); \theta^a), \vec{\Phi}^a(\vec{x}_{ijk}, 0) = \vec{x}_{ijk}; \end{aligned} \quad (10)$$

$$\begin{aligned} \vec{\Phi}^b(\vec{x}_{ijk}, t_{m+1}) &= \vec{\Phi}^b(\vec{x}_{ijk}, t_m) \\ &+ \tau \vec{V}(\vec{\Phi}^b(\vec{x}_{ijk}, t_m); \theta^b), \vec{\Phi}^b(\vec{x}_{ijk}, 0) = \vec{x}_{ijk}. \end{aligned} \quad (11)$$

The generated maps for each partition, $\hat{f}_1^a$ and $\hat{f}_1^b$, are obtained as follows:

$$\hat{f}_1^a(\Phi^a(\vec{x}_{ijk}, 1)) = f_0^a(\vec{x}_{ijk}), \quad \hat{f}_1^b(\Phi^b(\vec{x}_{ijk}, 1)) = f_0^b(\vec{x}_{ijk}). \quad (12)$$

For employing rigidity constraints, the loss function is modified to consist of two components: the squared Wasserstein distance between the combined deformed map $\hat{f}_1^a + \hat{f}_1^b$ and the target density map f_1 , and the local rigidity constraint $L_{\text{rigid}}(\cdot)$ on the two velocity fields²⁰ (Supplementary Note X). Consequently, the total loss of the model is

$$\mathcal{W}_2(\hat{f}_1^a + \hat{f}_1^b, f_1)^2 + \lambda^a L_{\text{rigid}}(\vec{V}(\theta^a)) + \lambda^b L_{\text{rigid}}(\vec{V}(\theta^b)). \quad (13)$$

Here, λ^a and λ^b are hyperparameters of the model that control the rigidity degree of the velocity field. The selection of λ^a and λ^b , as well as the partitioning process, are described in detail in Supplementary Note XI. The model is trained using ADAM optimization with the same parameters as before. After obtaining the optimal parameters θ^{a*} and θ^{b*} , the final optimal trajectory can be generated as follows:

$$\{f_0, \hat{f}_{t_1}^a + \hat{f}_{t_1}^b, \dots, \hat{f}_{t_M}^a + \hat{f}_{t_M}^b\}, \quad (14)$$

where

$$\hat{f}_{t_m}^a = f_0((\Phi^a)^{-1}(\cdot, t_m)), \quad \hat{f}_{t_m}^b = f_0((\Phi^b)^{-1}(\cdot, t_m)). \quad (15)$$

Metrics for density map quality

The model-to-map FSC is used to measure the similarity between the CryoTRANS-generated density and the atomic model. The unmasked FSC curves are depicted in this study. This is because CryoTRANS requires density maps to be represented as probability distributions, necessitating that each voxel's value be positive. During the preprocessing stage, any negative values in the density map are reset to zero. This process effectively acts as a mask applied to the density map itself.

The Q-score of the CryoTRANS-generated density map is compared with the Q-score of the target map. To further evaluate the accuracy of CryoTRANS, map-to-model in Phenix is used to build new atomic models from both the generated map and the target map, respectively. Subsequently, the TM-score between the reconstructed model from Phenix and the corresponding reference model from the PDB bank is calculated, serving as an additional measure of structure accuracy. Chain-comparison in Phenix is also used to compute the backbone coverage, providing further insights into the accuracy of the generated structures.

Statistics and reproducibility

The model is described in detail through mathematical formulation, and both the Python code and raw data are uploaded. The experiments mentioned in the article can be reproduced using the provided code and data. The neural network of CryoTRANS employs random initialization, so the results of experiments will exhibit some randomness, which may cause fluctuations in the model-to-map FSC curves of the generated densities. The displayed results are based on the best outcomes from three repeated experiments.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All source data underlying the graphs and charts in the main figures are available in Supplementary Data 1. The datasets analyzed in this study were downloaded from the EMPIAR repository. Atomic models are obtainable using the Protein Data Bank (PDB) ID mentioned in the text. Experimental results, including trajectories, of the datasets incorporated in this study have been deposited at <https://github.com/mxhulab/cryotrans-demos>.

Code availability

CryoTRANS is open source and released on GitHub with its homepage available at <https://github.com/mxhulab/cryotrans>.

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Author contributions

C.B., M.H., and Z.S. initiated the project. C.B., M.H., Z.S., X.F., Q.Z., and L.J. clarified the mathematical principles. X.F., Q.Z., M.H., and C.B. developed CryoTRANS and carried out testing. H.Z. provided support in analyzing the A2M dataset. X.F., Q.Z., M.H., and C.B. analyzed the data. M.H., C.B., Z.S., X.F., and J.Z. wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Zuoqiang Shi, Mingxu Hu or Chenglong Bao.

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